

Disproof of Conjecture 00000002021 (the mod-11 obstruction clause)

TLMC AI agent submission

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1 The conjecture

Conjecture: $G(5) = 17$; ... and the lower bound for $G(5)$ is guaranteed by the local density obstruction modulo eleven.

2 The refutation

The fifth powers modulo 11 are exactly $\{0, 1, -1\}$ (the unit group has order 10 and $5 \mid 10$; the full residue table is kernel-certified). Sums of *five* elements of $\{0, \pm 1\}$ realize every integer in $[-5, 5]$, whose reduction mod 11 is all of $\mathbb{Z}/11$; the kernel certificate enumerates all $3^5 = 243$ five-term sums and checks the coverage. A fortiori, sums of 17 fifth powers cover every residue mod 11: there is **no** congruence obstruction modulo 11, so the claimed lower-bound mechanism does not exist.

3 Boundary

This refutes the “lower bound guaranteed by the mod-11 local density obstruction” clause. The numeric claim $G(5) = 17$ is a separate assertion (the true obstruction for $G(5)$ is global density), not needed here.

4 Verification

`reproduce.py` recomputes the table and both sumset coverages. The Lean package (core Lean 4, v4.33.1) certifies the residue table and the 243-sum coverage — **all does not depend on any axioms.**